

TNBC–BIR state decomposition

Literature–anchored locked signatures → context–specific evidence map

Locked modules

epithelial/plasticity
APC/CXCL9/HLA–II
suppressive myeloid
proliferation

Bulk evidence

higher pCR odds in
I–SPY2 context
GSE25066 negative
chemotherapy boundary

Source support

myeloid/DC
macrophage/APC
B–cell/APC
public scRNA

Interpretation: treatment–context state mapping, not universal pCR prediction